

between a positive (called the cathode) and a negative electrode (called the anode). A fuel cell also has channels on each side, carrying a fuel (usually hydrogen or methanol) over the anode, and oxygen or air over the cathode. That enables fuel cells to keep producing electricity indefinitely, as long as there is a supply of fuel and air.

In the fuel cell, the electrolyte membrane keeps the fuel on one side of the cell from migrating across to the other side. Such migration contaminates the cell and can lead to a significant drop in efficiency. One big advantage of the new membranes produced by the MIT is that they are especially good at blocking the migration of methanol fuel.

Control your mobile with eyes

In your wildest imagination, have you ever thought of making calls from your mobile phone using your eye movements? Sounds fantastic? If the prototype of a new mobile phone unveiled by NTT DoCoMo makes it into mainstream mobile technology, your fantasy may soon be turned into reality.

At the *Mobile World Congress* held in Barcelona, Japanese mobile phone operator NTT DoCoMo demonstrated a prototype mobile phone that detects eye movement and translates it into on-screen interactions. The prototype device has special electrodes—attached

to the earphone—that can pick up the movement of the eye. Eyes have an electrical potential—positive at the cornea and negative at the retina—that changes depending on the movement of the eyeball. The earphone electrodes are able to read these changing currents and the mobile phone is pre-programmed to translate the information into a command.

Text-translating phone

The biggest challenge for tourists travelling to foreign countries is overcoming the language barrier. Very often, the best they can do is resort to sign language, which is not a viable solution always. Now, Internet giant Google is reportedly engaged in a software that translates the text captured by a phone camera.

Google demonstrated the technology recently at the *Mobile World Congress*. An engineer took a shot of a German dinner menu with a mobile phone powered by Google's Android software. The information was then routed to Google's servers via an inbuilt phone application, which sent a translation back to the phone.

Some phones already have an application that translates text from pictures, but mostly the processing is done by the phone itself. Google claims to be the first to send the information to its servers for processing. It can thus apply the power of its computers for

faster, more accurate results.

New solar-powered phone

A new phone by Puma has a 7cm (2.8-inch) touchscreen and a large solar panel installed on its back, so the battery can be recharged on the move. It also features a 3.2-megapixel camera, built-in mapping software and Internet access.

For the phone, Puma has developed a special operating system along with Sagem, and offers a pedometer, stopwatch and GPS tracker, so it's very handy for users to see the route they've run or cycled. They can also download additional applications and widgets from the Puma website. The phone will be available for sale in April.

Cellphone for the elderly

Older people will now be able to hold their own against tech-savvy youngsters in the technology rat race. A Sagem device, called *CosyPhone*, has a large keypad with separated buttons and uses near-field communication technology for older users to keep in touch with family and friends without the trouble of scrolling through menus. This phone makes use of cards, which can be customised with phone numbers of friends and relatives or even their photos. To connect, the user simply has to 'wave' his phone over that card to speak to that person or send an SMS. The device uses a recharger cradle

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